Capítulo 3

The influence of phylogenetic diversity on decomposition, respiration, and productivity in multitrophic freshwater ecosystems[‡]

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Abstract

Biodiversity loss is a serious issue worldwide. The consequences of species loss are usually drastic, since their functions in ecosystems could also possibly being lost. The consequences of the lost of primary producers for terrestrial ecosystem functioning have been studied in length, but the effects of other facets of biodiversity, such as phylogenetic diversity on multitrophic aquatic ecosystems are still poorly understood. Phylogenetic diversity can be a good proxy for unmeasured traits of species belonging to distinct trophic groups. Here, we re-analyzed experimental data that manipulated species richness and composition independently in multitrophic freshwater ecosystems. We tested the effect of two richness-independent phylodiversity measures (MNTD and MPD) on ecosystem productivity, respiration, and decomposition. We found that the linear term MNTD affected decomposition and respiration, while productivity was affected by the quadratic term of MNTD. On the other hand, neither phylodiversity measure significantly predicted productivity and respiration. Our results present new perspectives for the understanding of the effects of species loss to multitrophic freshwater ecosystems.

Key-words: ecophylogenetics, food webs, niche, effect traits, biodiversity loss.

Introduction

We are currently facing a massive biodiversity loss on Earth, which parallels big extinctions of the geological history of our planet (Barnosky et al., 2011). This biodiversity crisis poses key questions: what are the consequences of biodiversity loss for the functioning of ecosystems? Which ecosystem services will be most affected by the loss of species or functional groups (Loreau, 2010; Naem et al., 2012)? Historically, field experiments have manipulated the number of species and functional groups of plants to investigate their impact on ecosystem functions (EF), such as productivity (reviewed in Loreau, 2010; Cardinale et al., 2011). Nonetheless, the experimental designs have been criticized for being too simplistic by not including vertical diversity, that is, more than one trophic level (Duffy, 2009).

There is comparatively less work in freshwater ecosystems (Petchey et al., 2002, Giller et al. 2004), than in their terrestrial counterparts, especially involving multitrophic systems and measuring multiple ecosystem functions (Duffy et al., 2007, Woodward 2009). Trophic diversity and food web dynamics have recently been incorporated into ecosystem functioning experiments (Thebault and Loreau, 2003; 2006; Ives et al., 2005; Loreau, 2010), especially in freshwater systems (e.g., Cardinale et al., 2006; Duffy et al., 2007; Woodward, 2009).

Previous works on the effects of vertical diversity on EF suggest that differences in resource acquisition by animals and plants would produce more complex results than those predicted by experiments that manipulated only single trophic levels (e.g., producers; Duffy et al. 2007). For example, trophic chain length is predicted to influence the likelihood of trophic cascades (Duffy et al. 2007) and indirect effects (Downing and Leibold 2002), which can influence density-dependent relationships between predators and prey, and prey biomass. Also, prey edibility can change their dominance patterns and play a key role in mediating selection effects (Thebault and

Loreau 2003; Duffy et al. 2007). Accordingly, a model suggests (Thebault and Loreau 2003) that productivity can have a unimodal response to increasing levels of producer richness, as a consequence of the variation in prey edibility and predator specialization (a measure of food web connectivity). The predictions of the model were partially corroborated by a microcosm experiment (Filip et al., 2014), since these authors found a decrease in prey biovolume in the presence of specialist consumers, rather than generalist, as predicted by Thebault & Loreau's model. Yet, this result was largely driven by two key traits: growth rate of preys and consumer grazing rate. However, such experiments and theoretical models manipulated species richness assuming that species represented independent entities and would adequately represent the full niche space occupied by species of a community (Loreau 2010). Other nonlinear or even negative effects of diversity are expected at high diversity, due to overexploitation of resources by consumers (Loreau 2010; chapter 4).

Concurrently, there is growing evidence that facets of biodiversity, other than taxonomic diversity, are involved in regulating ecosystem processes, such as functional and phylogenetic diversities (e.g., Srivastava et al., 2012; Cadotte, 2013). A functional trait can be any characteristic measured in the individual level that potentially affects fitness and that could indicate how species interact with each other, where they live, and which resources they use (McGill et al., 2006). Ultimately, traits can be surrogates for describing the niche space of a community and therefore how efficiently resources are used. Accordingly, traits affecting a given ecosystem function are called effect traits (Viole et al., 2009). These traits are key elements to understand the link between diversity and ecosystem functioning (Loreau, 2010). However, it is hard to define a priori which and how many traits are involved in ecosystem functioning (Cadotte, 2013). Defining effect traits can be an even more challenging task when dealing with

multitrophic systems. In such scenarios with distant species, a phylogeny may be a better proxy, since it emphasizes similarity within functional groups (which are close relatives), in contrast with species from different functional groups.

Conversely, phylodiversity measures can be used as a proxy for a set of unmeasured traits, and thus provide a better representation of trait space, given that those traits are correlated with the phylogeny, i.e., have a phylogenetic signal (Srivastava et al., 2012). Recent experiments manipulating a single trophic level demonstrated that distant relatives could contribute more to biomass than recently diverged species pairs (Venail et al., 2008; Cadotte et al. 2008; Venail and Vives, 2013), which would be driven by complementary and efficient use of available resources (Cadotte et al., 2008; Venail and Vives, 2013).

However, phylodiversity measures may not be good predictors of ecosystem functioning in some situations, for example, when effect traits that drive a certain ecosystem functioning do not have phylogenetic signal at the community scale (Srivastava et al., 2012). Accordingly, the existence of phylogenetic signal also depends on community phylogenetic structure (e.g., Graham et al., 2012), environmental harshness (Burns and Strauss, 2012), and spatial scale (Kembel and Cahil, 2011). Consequently, the presence of phylogenetic signal in the regional phylogeny does not guarantee that there will be a signal at the finer spatial scale of a community or experimental plots (Srivastava et al., 2012). Additionally, environmental conditions that do not favor strong trait conservatism (e.g., Burns and Strauss, 2012) or that promote strong selection due to interspecific competition (Gravel et al., 2012) may result in trait overdispersion, which undermine the effect of phylodiversity measures on EF. Nonetheless, no study so far investigated experimentally the role of

phylodiversity on the functioning of freshwater ecosystems with multiple trophic levels (see Thompson et al. 2015 for a non-experimental approach).

Here, we re-analyzed the results of an experiment that investigated the effect of species richness and composition of multitrophic freshwater communities on multiple ecosystem functions (Downing and Leibold, 2002; 2010). Assuming that measures of phylogenetic diversity represent the niche space of a community better than taxonomic richness, following a theoretical model (Thebault and Loreau, 2003), we hypothesize that food web dynamics and species interaction may influence the effect of phylogenetic diversity on ecosystem function in a nonlinear fashion (Loreau 2010, Mora et al. 2014). Unimodal and saturating relationships have been found in previous studies using either taxonomic richness or phylodiversity (reviewed in Loreau 2010), but linear responses are also common (e.g., Downing and Leibold 2002). Despite not being designed explicitly to manipulate measures of phylogenetic diversity this experiment offers an opportunity to investigate for the first time how well phylodiversity measures predict ecosystem functioning in multitrophic aquatic ecosystems.

Material and Methods

Experimental design

We re-analyzed the data from a mesocosm experiment by Downing and Leibold (2002; 2010) that manipulated three levels of taxonomic richness with 3, 9, and 15 species containing equal numbers of three functional groups: macrophytes, periphyton grazers, and macroinvertebrate predators. Then, seven different species composition were nested within each richness level by randomly choosing one, three, or five species per functional group, comprising 21 different species composition. Each set of species

composition was replicated four times. The experiment lasted for 12 weeks, during which three ecosystem properties were measured every two weeks: productivity, respiration, and decomposition (Downing and Leibold, 2002). Productivity was measured as the net gain and respiration is the net loss of oxygen between dawn and dusk. Decomposition is the loss of dry mass per day of sugar maple leaves enclosed in mosquito netting.

Phylogeny building

Genes with low mutation rates are necessary to recover the relationships between distantly related groups. Since no suitable set of such genes was available for all species in GenBank, we assembled a topological tree manually for the species pool used in the experiment, consisting of 19 species (we excluded *Vallisneria americana* and *Utricularia vulgaris* since all individuals of these species died during the experiment; Downing and Leibold, 2002). A pseudo-chronogram was then obtained with the BLADJ algorithm of the software Phylocom 4.2 (Webb et al., 2008) by evenly interpolating 8 nodes without known ages in between 10 nodes with known ages. The algorithm then returned a dated tree with branch lengths in millions of years (Fig. 1). Node ages were obtained from TimeTree (Hedges et al., 2015). A recent study (Cadotte, 2015) found that tree topology, i.e., correctly establishing species relationships, is far more important than branch length accuracy in predicting ecosystem functioning. Thus, a tree assembled by hand seems adequate for the goals of this paper.

Phylodiversity measures

We calculated two metrics of phylogenetic diversity for each of the 21 sets of species composition manipulated in the experiment: Mean Nearest Taxon Distance

(MNTD; Webb et al., 2002); and Mean Pairwise Distance (MPD; Webb et al., 2002), which are, respectively, the average distance between the closest-related species pairs in a community, and the average phylogenetic distance connecting species pairs in a community. Both measures are independent of species richness, but MPD captures deeper relationships in the phylogenetic tree, whereas MNTD captures relationships closer to the tips (Webb et al., 2002; Fig. S1). By using two measures of phylodiversity related to distinct depths of the phylogeny, we could investigate if the effect of phylogenetic relatedness on ecosystem function is more related to deep divergences or recent splits in the phylogeny, since we are dealing with three functional groups that diverged from each other long time ago. This approach was used in a recent study (Gravel et al., 2012), which found that productivity was influenced by the phylogenetic relationship between closely related bacteria, but not distantly diverged ones.

Data analysis

For each ecosystem function (productivity, respiration, and decomposition), we built two linear models with repeated measures for nested designs using linear and quadratic terms of either MNTD or MPD as predictors. Quadratic terms were included to test the prediction of nonlinear relationships between phylodiversity and ecosystem function. Since each ecosystem function was measured six times along the experiment, we included time as a repeated measure factor in the model. The experimental design included different species composition nested within each species richness treatment. Since phylodiversity measures were derived from species composition, they were included in the models as nested within each richness treatment.

Results

Simiarly to Downing and Leibold's experiment, taxonomic richness ($F_{2,492}$= 7.342; $P < 0.001$) significantly explained productivity, as did MPD ($F_{3,492}$=11.344; $P < 0.001$) and MNTD ($F_{3,492}$ = 18.634; $P < 0.001$; Table 1). Respiration was also predicted by MNTD, but not MPD (Table 1). Decomposition was the only function predicted by the quadratic term of a phylodiversity metric, in this case MNTD ($F_{1,492}$=4.26; P = 0.03), with this variable explaining about 43.5% of the variation in decomposition. Neither MPD nor time and richness explained significantly decomposition.

Discussion

We found that respiration, decomposition, and productivity were predicted by different phylodiversity measures that capture structure at distinct phylogenetic depths. Our results are similar to previous experiments that manipulated plant PD (Cadotte et al., 2008), and bacterial richness and MPD independently (Venail and Vives, 2013). These two experiments have found that either Faith's PD or MPD predicted ecosystem productivity, but manipulated only single trophic levels. Here, both MPD and MNTD influenced productivity in a multitrophic ecosystem.

MNTD seems to be a good predictor of decomposition, but its relationship is non linear. This result suggests that phylogenetic relationships among closely related species affects decomposition in a complex way. It also agrees with a meta-analysis (Srivastava et al. 2009) that found top-down effects driven by detritivores as more important than bottom-up forces in influencing decomposition in ecosystems. Other studies (e.g., Boyero et al. 2014) have suggested that changes in species identity of decomposers are a key element in driving the effect of diversity on decomposition rates, though the mechanisms are not fully understood, especially in aquatic ecosystems (see

Srivastava et al. 2009, Boyero et al. 2014). Our results further suggest that not only species identity, but also shallow phylogentic relationships influence decomposition rates. We found a positive, linear response of decomposition to MNTD (standardized partial coefficient = 0.108), which indicates that decomposition rates increases with increasing mean number of closely related species. There seems to be low support for selection effects as main drivers of ecosystem functioning in the experiment by Downing and Leibold (2002), since the effects of species are via indirect effects on other species. Therefore, it is tempting to suggest that the relationship we found between MNTD and decomposition is due to complimentary effects. A previous study (Gravel et al., 2012) found that productivity was better predicted by PD of ancestral lineages of bacteria, instead of evolved ones. Here, we have found the opposite pattern, indicating that changes in species phenotypes during a recent period of time have strong effects on decomposition rates (Harmon et al., 2009; Lundsgaard-Hansen et al., 2014). Therefore, the consequences of the loss of closely related species of detritivores could not be compensated by other species within trophic levels (see also Jabiol et al., 2013; Gessner et al., 2010). All insects used in the experiment were from lentic habitats, whose detrital processing ability is different from those of lotic environments, suggesting other groups of organisms should play important roles in decomposing plant material.

Productivity and respiration were both predicted by MNTD, while MPD predicted productivity alone. It is not clear what mechanisms are involved in the influence of evolutionary history on productivity, but previous studies with grassland plants have equated PD with niche differentiation and the complementarity effect (Venail et al., 2008; Cadotte, 2013). Accordingly, a recent study (Turnbull et al., 2013) pointed out that, under some circumstances, stabilizing niche differences could be

related to the complementarity effect (but see discussion in Carroll et al. 2011 and Loreau et al., 2012). Consequently, it seems that food web dynamics (e.g., indirect effects) interact with shallow phylogenetic history to affect respiration, while both shallow and deep phylogenetic relationships influence productivity via complementary effects (i.e., niche partitioning). Our results also add to those of Downing & Leibold (2002), who found species composition affected respiration. By considering species relationships, we found that the effects of species compotision on respiration is likely due to shallow phylogenetic relationships.

In conclusion, our results suggest the effect of phylogenetic history on decomposition is related to recent splits in the phylogeny. However, much remains to be done to uncover the mechanism driving the relationship between phylodiversity and ecosystem functioning. For example, effect traits and those related to competitive dominance in a community need to be analyzed along with the phylogeny in order to test how they evolved along the phylogeny (e.g., Gravel et al., 2012) and how these patterns influence their effects on ecosystem functioning.

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Figure legends

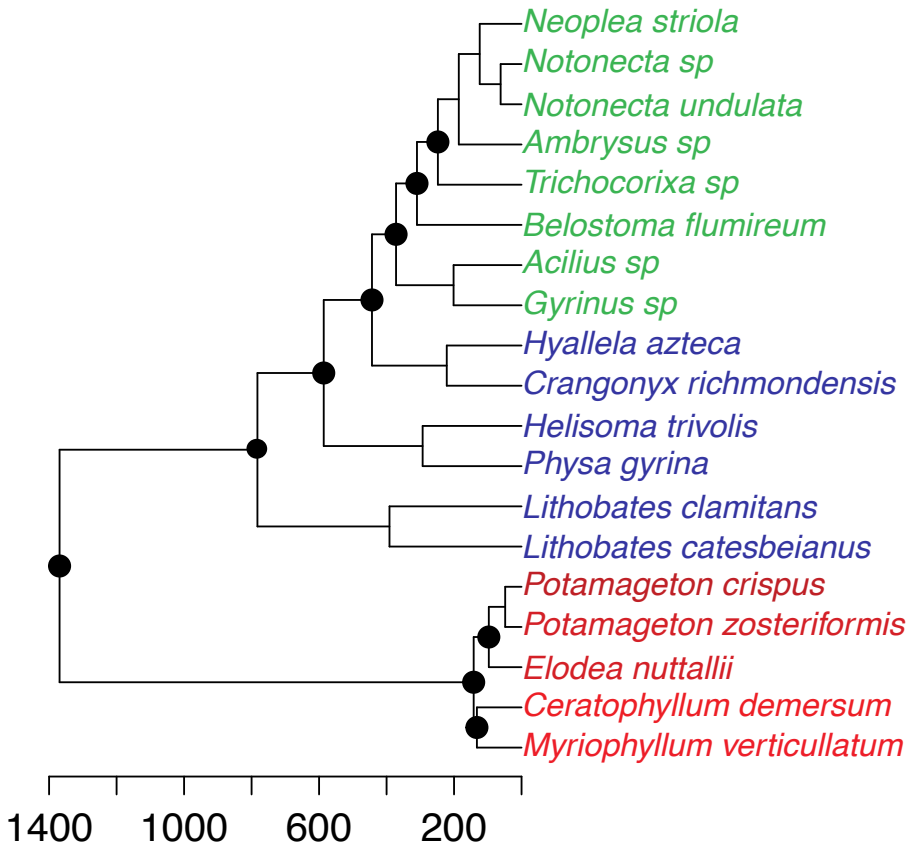


Figure 1. Pseudo-chronogram assembled for the species used in the Downing and Leibold's (2002) experiment, showing the nodes (black dots) with known ages (in Millions of years) used to calibrate the chronogram. Species in red are macrophytes, periphyton grazers are in blue, and invertebrate predators are colored in green.

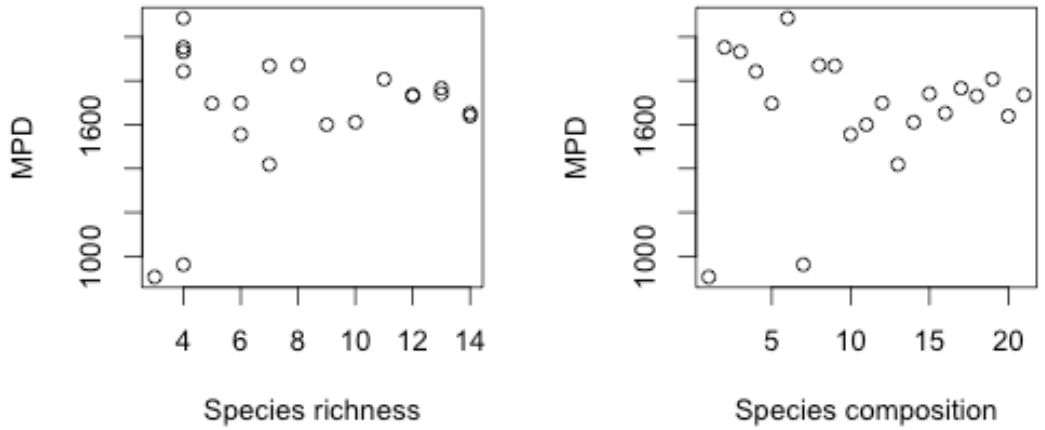


Figure S1. Variation in MPD across species richness and composition, nested within each species richness treatment. Notice the independence between MPD and species richness and composition.

Table legend

Table 1. Results of rm-ANOVA for nested designs for productivity, respiration, and decomposition (response variables) as a function of MPD, MNTD, and richness (factor with three levels). Response variables were measured six times throughout the experiment.

		DF	Sum Sq	Mean Sq	F value	*P* value
Productivity	Richness	2	0.047	0.023	7.342	**<0.001**
	MNTD	3	0.180	0.060	18.634	**<0.001**
	$MNTD^2$	1	0.005	0.005	1.726	0.189
	Time	5	0.028	0.005	1.797	0.111

	Residuals	492	1.586	0.003		
	Richness	2	0.047	0.023	7.048	**<0.001**
	MPD	3	0.114	0.038	11.344	**<0.001**
	MPD^2	1	0.003	0.003	0.918	0.338
	Time	5	0.031	0.006	1.857	0.010
	Residuals	492	1.653	0.003		
Respiration	Richness	2	0.023	0.011	5.922	**0.002**
	MNTD	3	0.058	0.019	9.957	**<0.001**
	$MNTD^2$	1	0.000	0.000	0.335	0.562
	Time	5	0.010	0.002	1.02	0.402
	Residuals	492	0.968	0.001		
	Richness	2	0.023	0.011	5.677	**0.003**
	MPD	3	0.012	0.004	2.047	0.169
	MPD^2	1	0.003	0.003	1.895	0.169
	Time	5	0.011	0.002	1.098	0.360
	Residuals	492	1.010	0.002		
Decomposition	Richness	2	0.16	0.078	0.083	0.919
	$MNTD^2$	1	4.26	4.264	4.541	**0.033**
	Time	5	5.15	1.029	1.096	0.361
	MNTD	3	2.17	0.723	0.770	0.510
	Residuals	492	461.97	0.939		
	Richness	2	0.16	0.078	0.083	0.920
	MPD	3	1.51	0.504	0.532	0.659
	MPD^2	1	2.29	2.289	2.420	0.920
	Time	5	4.39	0.878	0.928	0.463
	Residuals	492	465.37	0.945		